

KKT-free Constrained Optimization and Application to River-Flow Modeling *

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Abstract

This paper explores Augmented Lagrangian methods for derivative-free continuous optimization under constraints. The theoretical analysis proceeds without assuming the existence of derivatives, and the proofs of these results bypass traditional KKT conditions. Numerical experiments demonstrate the fitting of parameters within river flow models. These models address the estimation of hydraulic parameters by minimizing their dispersion, subject to both differential equations and empirical observations. The experiments are situated within the framework of the CRIAB project, which focuses on the study of dam failure.

Key words: River flow modeling, Augmented Lagrangian, convergence, derivative-free optimization.

1 Introduction

Constrained optimization algorithms that do not use derivatives have been a fruitful area of research, driven by their applicability to a wide range of problems in engineering and applied sciences. See [41] for a recent application to fitting parameters of Climate Models. Augmented Lagrangian methods without derivatives of the objective function and constraints have been analyzed in the literature. See [6, 3, 24, 35, 39, 40, 49], among others. Most Augmented Lagrangian algorithms involve solving a single constrained (typically box-constrained) problem at each iteration. Consequently, their efficiency is closely tied to the efficiency of single-constraint derivative-free algorithms [4, 5, 3, 18, 14, 13, 23, 25, 35, 36, 38, 42, 43, 44, 54, 55].

The fact that the global convergence theory of the Augmented Lagrangian method is independent of the smoothness of the objective function or the constraints [12] is the main motivation for developing a convergence theory in which the subproblem solvers do not use derivatives.

Given any penalty method, in which subproblems are solved with an increasing degree of penalty for infeasibility, the Augmented Lagrangian modification consists of penalizing, not with respect to the true constraints, but with respect to a shift of them, with the objective of avoiding the use of large penalties.

Notably, the mere strategy for manipulating the penalty parameter and the shifts produces interesting results that are independent of the method used to solve the subproblems. That is the point of view adopted in this paper, in which results are proven that are adaptable to the most diverse subproblem solvers, from the most naive to the most sophisticated. This approach appears to be missing from the existing literature on Augmented Lagrangians, despite its apparent necessity.

This work introduces a general framework for the Augmented Lagrangian approach, allowing for an arbitrary choice of 'approximate solution' for the subproblems. The theoretical analysis examines the

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properties of the resulting algorithm under this general assumption. This approach diverges from most derivative-free Augmented Lagrangian methods in the literature which, despite avoiding derivatives in computations, typically rely on their existence and continuity to establish KKT-like convergence properties.

The numerical experiments presented in this paper focus on estimating parameters for river-flow models, motivated by the activities of the CRIAB group. This group, established in 2019 at the University of Campinas, aims to develop mathematical models for natural disaster prevention, in response to the devastating collapse of the Brumadinho dam in Minas Gerais, Brazil. The CRIAB group comprises members from various Latin American universities and institutions.

In addition to addressing a critical problem in natural disaster prevention, our experiments also serve as a testbed for the new implementation of Powell’s software BOBYQA [54], recently introduced in [55].

This work introduces a new formulation for estimating friction coefficients in hydraulic models. The proposed approach accounts for the spatial variability of the coefficients, leading to a constrained optimization problem that maintains data fidelity while minimizing parameter dispersion. This methodology represents a novel contribution to the field of Manning coefficient estimation for real river systems.

The remainder of this paper is organized as follows. Section 2 presents the general algorithm and establishes the theoretical results. Section 3 describes the parameter-fitting problem for natural channel models, while Section 4 reports the corresponding numerical results. Finally, Section 5 draws conclusions and suggests directions for future research.

Notation

- The symbol $\|\cdot\|$ denotes an arbitrary vector norm.
- $K_1 \subseteq K_2$ means that both K_1 and K_2 are sequences of natural numbers and K_1 is a subsequence of K_2 .
- For all $y \in \mathbb{R}^n$, $y_+ = (\max\{0, y_1\}, \dots, \max\{0, y_n\})^T$.
- $\mathbb{N} = \{0, 1, 2, \dots\}$.

2 Derivative-free Safeguarded Augmented Lagrangian

The form of the problems addressed in this paper is as follows:

$$\text{Minimize } f(x) \text{ subject to } h(x) = 0, g(x) \leq 0, x \in \Omega, \quad (1)$$

where $\Omega \subset \mathbb{R}^n$ is a compact set. We assume that the functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and $g : \mathbb{R}^n \rightarrow \mathbb{R}^p$ are continuous. First derivatives are not assumed to exist anywhere.

Given $\rho > 0$, $\lambda \in \mathbb{R}^m$, $\mu \in \mathbb{R}_+^p$, and $x \in \mathbb{R}^n$, the definition of the Augmented Lagrangian function $L_\rho(x, \lambda, \mu)$ is:

$$L_\rho(x, \lambda, \mu) = f(x) + \frac{\rho}{2} \left\{ \sum_{j=1}^m \left[h_j(x) + \frac{\lambda_j}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(x) + \frac{\mu_i}{\rho} \right]_+^2 \right\}. \quad (2)$$

The definition (2) corresponds to the well-known PHR (Powell-Hestenes-Rockafellar) [31, 53, 58] classical Augmented Lagrangian formulation. See, also, [11, 12, 16, 17, 32].

The idea is that the best strategy to minimize a function subject to constraints may not be to minimize the objective function plus a large penalty for constraint violations, but rather to minimize the objective function plus a moderate penalty for suitably shifted constraints. By properly choosing the shifts λ_j/ρ and μ_i/ρ , it is likely that very large penalty parameters ρ will not be necessary. Notably, the plausibility of this idea does not depend on the differentiability of the objective function or the constraints.

Algorithm 2.1 is a minimalist version of the Safeguarded Augmented Lagrangian method (Algorithm 4.1 of [12]). The way in which $x^k \in \Omega$ is computed at every iteration of the method is not specified at all. In practice, x^k will be usually obtained by means of some minimization criterion with respect to the augmented Lagrangian.

Algorithm 2.1

Let $0 < \tau < 1$, $\gamma > 1$, and $\lambda_{min} < \lambda_{max}, \mu_{max} > 0$. At the first outer iteration the penalty parameter $\rho_1 > 0$ and the safeguarded Lagrange multipliers estimates $\lambda^1 \in \mathbb{R}^m$ and $\mu^1 \in \mathbb{R}^p$ satisfy:

$$\lambda^1 \in [\lambda_{min}, \lambda_{max}]^m \text{ and } \mu^1 \in [0, \mu_{max}]^p.$$

Initialize $k \leftarrow 1$.

Step 1. Compute $x^k \in \Omega$.

Step 2. Define, for all $i = 1, \dots, p$,

$$V_i^k = \min \left\{ -g_i(x^k), \frac{\mu_i^k}{\rho_k} \right\},$$

If $k = 1$ or

$$\max\{\|h(x^k)\|_\infty, \|V^k\|_\infty\} \leq \tau \max\{\|h(x^{k-1})\|_\infty, \|V^{k-1}\|_\infty\}, \quad (3)$$

define $\rho_{k+1} = \rho_k$. Else, define $\rho_{k+1} = \gamma\rho_k$.

Step 3. Define $\lambda^{k+1} \in [\lambda_{min}, \lambda_{max}]^m$ and $\mu^{k+1} \in [0, \mu_{max}]^p$.

Step 4. Set $k \leftarrow k + 1$ and go to Step 1.

The description of Algorithm 2.1 does not mention the Augmented Lagrangian function at all. In practice, x^k is computed as an approximate minimizer of the subproblem

$$\text{Minimize } L_{\rho_k}(x, \lambda^k, \mu^k) \text{ subject to } x \in \Omega. \quad (4)$$

However, theoretical results will not rely on this characterization. In this sense, Algorithm 2.1 is a minimalist version of Algorithm 4.1 of [12].

For all $k \in \mathbb{N}$ $N(x^k)$ will be the set of points $z \in \Omega$ such that

$$L_{\rho_k}(x^k, \lambda^k, \mu^k) \leq L_{\rho_k}(z, \lambda^k, \mu^k). \quad (5)$$

The points of $N(x^k)$ will be called “neighbors” of x^k . By extension, if x^* is a limit point of $\{x^k\}$, the set of neighbors $N(x^*)$ will be defined by

$$z \in N(x^*) \text{ if and only if there exists } K \subseteq \mathbb{N} \text{ and } z^k \in N(x^k) \text{ such that } \lim_{k \in K} z^k = z.$$

Remarks

- Note that the algorithm does not require the explicit computation of $N(x^k)$. If a global minimizer x^k of $L_{\rho_k}(x, \lambda^k, \mu^k)$ is obtained, then $N(x^k) = \Omega$.
- In Algorithm 2.1, the set Ω is quite general, accommodating complicated nonlinear constraints and even integer constraints. Consequently, the computation of x^k can be performed using alternative constrained Derivative-Free algorithms, such as those introduced in [3, 14, 13, 26, 28, 42, 43, 51].

The following theorem proves that, if a subsequence of $\{x^k\}$ converges to x^* and $\{\rho_k\}$ is bounded, x^* is a feasible point. If ρ_k is not bounded and a sequence of points $z^k \in N(x^k)$ converges to z then x^* is at least as feasible as z . Therefore, if x^k is a global minimizer of $L_{\rho_k}(x, \lambda^k, \mu^k)$ (so, $N(x^k) = \Omega$), it turns

out that x^* is a global minimizer of the sum of squares of infeasibilities. In this sense, Theorem 2.1 is a generalization of Theorem 5.1 of [12].

Since Ω is compact, limit points of the sequence $\{x^k\}$ generated by Algorithm 2.1 necessarily exist. Theorem 2.1 proves that limit points are feasible provided that the sequence of penalty parameters is bounded. So, boundedness of the penalty parameters implies that the feasible region is non-empty. In general, it is not possible to prove that penalty parameters are bounded, as the existence of feasible points cannot be guaranteed. In Augmented Lagrangian theories boundedness of penalty parameters is guaranteed under rather restrictive conditions that involve feasibility, differentiability of the functions, and regularity assumptions [12]. According to Theorem 5.1 of [12], even when the penalty parameters are unbounded, limit points are global minimizers of the infeasibility measure $\|h(x)\|_2^2 + \|g(x)_+\|_2^2$. However, this result relies on the assumption that the iterates x^k are global minimizers of the corresponding Augmented Lagrangian functions. The present work only assumes that the Augmented Lagrangian at x^k is no bigger than the Augmented Lagrangian at the points $N(x^k)$. Consequently, as ρ_k tends to infinity, the convergence analysis is limited to establishing that the infeasibility measure at a limit point x^* does not exceed that of its neighbors $N(x^*)$. In other words, x^* is at least as feasible as any of its neighbors.

Note that, although boundedness of the penalty parameters implies feasibility, the converse is not true. Feasible limit points may exist even in the case that the penalty parameters are unbounded. Unboundedness of the penalty parameters could be forced by taking a very small τ in the test (3). Of course, this is not recommendable.

Theorem 2.1 *Assume that $\{x^k\}$ is a sequence generated by Algorithm 2.1. Let $K \subseteq \mathbb{N}$ be such that*

$$\lim_{k \in K} x^k = x^*. \quad (6)$$

Then, if $\{\rho_k\}$ is bounded, we have that $h(x^) = 0$ and $g(x^*) \leq 0$.*

Otherwise, if a sequence $\{z^k\} \subset \Omega$ exists such that

$$\lim_{k \in K} z^k = z \quad (7)$$

and $z^k \in N(x^k)$ for all $k \in K$, we have that $z \in \Omega$ and

$$\|h(x^*)\|_2^2 + \|g(x^*)_+\|_2^2 \leq \|h(z)\|_2^2 + \|g(z)_+\|_2^2. \quad (8)$$

Proof Since Ω is compact, by (6), we have that $x^* \in \Omega$.

Assume, firstly, that $\{\rho_k\}$ is bounded. By Step 4, there exists k_0 such that $\rho_k = \rho_{k_0}$ for all $k \geq k_0$. Therefore,

$$\max\{\|h(x^k)\|_\infty, \|V^k\|_\infty\} \leq \tau \max\{\|h(x^{k-1})\|_\infty, \|V^{k-1}\|_\infty\}$$

for all $k \geq k_0$. Then,

$$\lim_{k \rightarrow \infty} \|h(x^k)\| = \lim_{k \rightarrow \infty} \|V^k\| = 0. \quad (9)$$

So, for all $i = 1, \dots, p$,

$$\lim_{k \rightarrow \infty} \left| \min \left\{ -g_i(x^k), \frac{\mu_i^k}{\rho_k} \right\} \right| = 0.$$

Since $\mu_i^k \geq 0$ this implies that

$$\lim_{k \rightarrow \infty} g_i(x^k)_+ = 0. \quad (10)$$

By (9) and (10) we have that $\|h(x^*)\| = 0$ and $\|g(x^*)_+\| = 0$.

Now suppose that $\{\rho_k\}$ is unbounded. By compactness, $z \in \Omega$.

By (5), for all $k \in K$ we have that

$$L_{\rho_k}(x^k, \lambda^k, \mu^k) \leq L_{\rho_k}(z^k, \lambda^k, \mu^k). \quad (11)$$

Then, by the definition of the Augmented Lagrangian, for all $k \in K$ we have that

$$f(x^k) + \frac{\rho_k}{2} \left\{ \sum_{j=1}^m \left[h_j(x^k) + \frac{\lambda_j^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \leq f(z^k) + \frac{\rho_k}{2} \left\{ \sum_{j=1}^m \left[h_j(z^k) + \frac{\lambda_j^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(z^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\}.$$

So,

$$\frac{1}{\rho_k} f(x^k) + \frac{1}{2} \left\{ \sum_{j=1}^m \left[h_j(x^k) + \frac{\lambda_j^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \leq \frac{1}{\rho_k} f(z^k) + \frac{1}{2} \left\{ \sum_{j=1}^m \left[h_j(z^k) + \frac{\lambda_j^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(z^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\}.$$

Taking limits for $k \in K$ and using that λ^k and μ^k are bounded, $\rho_k \rightarrow \infty$, and the continuity of h, g , and $\max$, we obtain:

$$\sum_{j=1}^m h_j(x^*)^2 + \sum_{i=1}^p g_i(x^*)_+^2 \leq \sum_{j=1}^m h_j(z)^2 + \sum_{i=1}^p g_i(z)_+^2.$$

So, (8) is proved. $\square$

Note that for proving this theorem it is essential that the shifts tend to zero when the penalty parameter tends to infinity. This property aligns with common sense, as imposing extreme penalization for infeasibility would render the introduction of shifts unnecessary.

The following Corollary expresses the thesis of Theorem 2.1 in a more compact way.

Corollary 2.1 *Assume that x^* is a limit point of a sequence generated by Algorithm 2.1. Then, x^* is a solution of*

$$\text{Minimize } \|h(z)\|_2^2 + \|g(z)_+\|_2^2 \text{ subject to } z \in N(x^*).$$

Moreover, if the sequence of penalty parameters is bounded, x^ is feasible.*

Proof: It follows immediately from Theorem 2.1. $\square$

Theorem 2.1 indicates that the likelihood of finding feasible points improves when the penalty parameters are bounded. However, the boundedness of these parameters is not under direct control, as their potential increase at each iteration is determined by the test (3). Properly updating the shifts λ^k/ρ_k and μ^k/ρ_k increases the chances of fulfilling of (3). The formulae

$$\lambda^{k+1} = \lambda^k + \rho_k h(x^k) \text{ and } \mu^{k+1} = (\mu^k + \rho_k g(x^k))_+, \quad (12)$$

subject to the safeguarded defined by λ_{min} , λ_{max} , and μ_{max} , are commonly used. A justification of these updates, without gradient-like arguments, may be found in [12, Section 4.1]. This update strategy plays a crucial role in balancing the penalty parameters and promoting convergence to feasible points.

Nevertheless, it is impossible to prove that the algorithm finds feasible points, as such points may not exist. Therefore, it is sensible to analyze the behavior of the algorithm's limit points, assuming feasibility. The following theorem explores this scenario. This theorem will assume boundedness of the penalty parameters, which, according to Theorem 2.1, is the sufficient condition that guarantees feasibility of any limit point.

In Theorem 2.2, it will be proved that, if the sequence of penalty parameters is bounded, the objective function value at any limit point is no bigger than the objective function value at any of its neighbors. The quality of this result is connected with the effectiveness of the algorithm employed for solving the subproblems. If this algorithm finds global minimizers of the Augmented Lagrangian function, the limit points are global minimizers of the constrained optimization problem (1).

Theorem 2.2 Assume that $\{x^k\}$ is a sequence generated by Algorithm 2.1 and that $\{\rho_k\}$ is bounded. Let $K \subseteq \mathbb{N}$ be such that

$$\lim_{k \in K} x^k = x^*. \quad (13)$$

Suppose that, for all $k \in K$, $z^k \in N(x^k)$ is such that

$$\lim_{k \in K} z^k = z, \quad (14)$$

and

$$h(z) = 0 \text{ and } g(z) \leq 0. \quad (15)$$

Then, $h(x^*) = 0, g(x^*) \leq 0$ and

$$f(x^*) \leq f(z).$$

Proof By Theorem 2.1, since $\{\rho_k\}$ is bounded, we have that

$$h(x^*) = 0 \text{ and } g(x^*) \leq 0.$$

By the definition of the Augmented Lagrangian function, for all $k \in K$ we have that

$$f(x^k) + \frac{\rho_k}{2} \left\{ \sum_{j=1}^m \left[h_j(x^k) + \frac{\lambda_j^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \leq f(z^k) + \frac{\rho_k}{2} \left\{ \sum_{j=1}^m \left[h_j(z^k) + \frac{\lambda_j^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(z^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\}.$$

Let $K_1 \subseteq K$ be such that $\rho_k \rightarrow \rho, \lambda^k \rightarrow \lambda^*, \mu^k \rightarrow \mu^*$ for $k \in K_1$. Then,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{j=1}^m \left[h_j(x^*) + \frac{\lambda_j^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \left\{ \sum_{j=1}^m \left[h_j(z) + \frac{\lambda_j^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(z) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\}.$$

Then,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{j=1}^m \left[\frac{\lambda_j^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \left\{ \sum_{j=1}^m \left[\frac{\lambda_j^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(z) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\}.$$

So,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{i=1}^p \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \left\{ \sum_{i=1}^p \left[g_i(z) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\}.$$

Thus,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{g_i(x^*) < 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 + \sum_{g_i(x^*) = 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \sum_{i=1}^p \left[\frac{\mu_i^*}{\rho} \right]^2.$$

If $g_i(x^*) = 0$ we have that $[g_i(x^*) + \mu_i^*/\rho]_+ = \mu_i^*/\rho$, then

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{g_i(x^*) < 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \sum_{g_i(x^*) < 0} \left[\frac{\mu_i^*}{\rho} \right]^2. \quad (16)$$

Finally, if $g_i(x^*) < 0$, by the continuity of g_i we have that $-g_i(x^k) > 0$ is bounded away from zero if $k \in K$ is large enough. But, by the boundedness of $\{\rho_k\}$ we have that $\lim_{k \in K} V_k = 0$. Then, by the definition of V_k , $\lim_{k \in K} \mu_i^k/\rho_k = 0$. So, $\mu_i^*/\rho = 0$. Then, by (16),

$$f(x^*) \leq f(x^*) + \frac{\rho}{2} \left\{ \sum_{g_i(x^*) < 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z),$$

as we wanted to prove. $\square$

Corollary 2.2 *Assume that x^* is a limit point of a sequence generated by Algorithm 2.1 and the sequence of penalty parameters is bounded. Then, x^* is a solution of*

$$\text{Minimize } f(z) \text{ subject to } z \in N(x^*), h(z) = 0, \text{ and } g(z) \leq 0.$$

Theorem 2.3 concerns optimality in the case that $\rho_k \rightarrow \infty$. As in Theorem 2.2, in Theorem 2.3 will prove that, under proper assumptions, the function value at a limit point of the Augmented Lagrangian algorithm is not smaller than the function value at its feasible neighbors.

Theorem 2.3 *Assume that $\{x^k\}$ is generated by Algorithm 2.1,*

$$\lim_{k \in K} x^k = x^*, z^k \in N(x^k), \lim_{k \in K} z^k = z \in N(x^*),$$

and $\rho_k \rightarrow \infty$. In addition, assume that

$$\lim_{k \in K} \|h(z^k)\| \sqrt{\rho_k} = 0, \text{ and } \lim_{k \in K} \|g(z^k)_+\| \sqrt{\rho_k} = 0. \quad (17)$$

Then, z is feasible and $f(x^) \leq f(z)$.*

Proof Since $\lim_{k \in K} z^k = z$, by the continuity of h and g and (17) we have that $h(z) = 0$ and $g(z) \leq 0$. So, z is feasible.

Since $z^k \in N(x^k)$,

$$\begin{aligned} f(x^k) &+ \frac{\rho_k}{2} \left\{ \sum_{j=1}^m \left[h_j(x^k) + \frac{\lambda_j^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \\ &\leq f(z^k) + \frac{1}{2} \left\{ \sum_{j=1}^m \left[\sqrt{\rho_k} h_j(z^k) + \frac{\lambda_j^k}{\sqrt{\rho_k}} \right]^2 + \sum_{i=1}^p \left[\sqrt{\rho_k} g_i(z^k) + \frac{\mu_i^k}{\sqrt{\rho_k}} \right]_+^2 \right\}. \end{aligned}$$

Then,

$$f(x^k) \leq f(z^k) + \frac{1}{2} \left\{ \sum_{j=1}^m \left[\sqrt{\rho_k} h_j(z^k) + \frac{\lambda_j^k}{\sqrt{\rho_k}} \right]^2 + \sum_{i=1}^p \left[\sqrt{\rho_k} g_i(z^k) + \frac{\mu_i^k}{\sqrt{\rho_k}} \right]_+^2 \right\}. \quad (18)$$

Taking limits in both sides of (18) we obtain the desired result. $\square$

Corollary 2.3 *Assume that x^* is a limit point of a sequence generated by Algorithm 2.1 and $\bar{N}(x^*) \subset N(x^*)$ is such that for all $z \in \bar{N}(x^*)$ and $z^k \in N(x^k)$ such that $z^k \rightarrow z$, (17) holds. Then,*

$$f(x^*) \leq f(z) \text{ for all } z \in \bar{N}(x^*) \text{ such that } h(z) = 0, \text{ and } g(z) \leq 0.$$

Corollary 2.4 *Assume that x^* is a limit point of a sequence generated by Algorithm 2.1 and $\tilde{N}(x^*) \subset N(x^*)$ is such that for all $z \in \tilde{N}(x^*)$ and $z^k \in N(x^k)$ such that $z^k \rightarrow z$, we have that z^k is feasible. Then,*

$$f(x^*) \leq f(z) \text{ for all } z \in \tilde{N}(x^*) \text{ such that } h(z) = 0, \text{ and } g(z) \leq 0.$$

Proof Note that the feasibility of z^k implies (17). Then, the proof follows from Corollary 2.3. $\square$

If x^* is feasible, z is a feasible neighbor of x^* , and $z^k \in N(x^k)$ is such that $z^k \rightarrow z$, we have, by continuity, that $\|h(z^k)\| \rightarrow 0$ and $\|g(z^k)_+\| \rightarrow 0$. In this case, the assumption (17) is prone to hold and it follows that x^* is better than its feasible neighbors. However, (17) can be guaranteed only under special circumstances. For example, (17) takes place if, for k large enough, $h(z^k) = 0$ and $g(z^k) \leq 0$. In practice, these observations yield algorithmic prescriptions regarding the stopping criteria of the subproblem solver. Namely, the subproblem solver should stop at x^k when $L(x^k, \lambda^k, \mu^k) \leq L(z, \lambda^k, \mu^k)$ holds for as many feasible or almost feasible points z as possible.

Theorems 2.1 and 2.2 have interesting consequences. Clearly, the behavior of $\{\rho_k\}$ plays an essential role. In fact, by Theorem 2.1, the boundedness of $\{\rho_k\}$ guarantees that a feasible point is found. This occurs when a value of ρ_k is reached beyond which merely shift modifications are enough to guarantee feasibility. Of course, this favorable situation is impossible if the problem (1) is infeasible. On the other hand, if (1) is feasible, penalty boundedness usually takes place when updates (12) are employed and x^k is computed by means of an adequate procedure.

If ρ_k tends to infinity the thesis of Theorem 2.1 is weaker, as it only says that the limit point x^* is at least as feasible as its neighbors. Nevertheless, this does not preclude x^* from being a feasible point.

For these reasons, since the goal is to find minimizers of (1), Theorem 2.2 deserves particular emphasis. This theorem specifies what happens to the objective function when the sufficient condition for feasibility is met. Specifically, Theorem 2.2 states that the objective function value at a limit point is less than or equal to the objective function value at its feasible neighbors. To maximize the impact of this result, it is desirable that the inequality $f(z) \geq f(x^k)$ holds for as many points $z \in \Omega$ as possible. A natural strategy, therefore, is to design algorithms where the cardinality of $N(x^k)$ is as large as possible. Although this may require excessive computational time, it is worth noting that the set $N(x^k)$ does not need to be explicitly computed.

For completeness, the following analysis addresses the convergence of Algorithm 2.1 in the smooth case. For this analysis the compact set Ω will be assumed to be a box:

$$\Omega = \{x \in \mathbb{R}^n \mid \ell \leq x \leq u\}, \quad (19)$$

where $\ell_i < u_i$ for all $i = 1, \dots, n$. Moreover, Step 1 of Algorithm 2.1 will be implemented in such a way that Assumption 2.1 holds.

Assumption 2.1

For all $k = 0, 1, 2, \dots$, there exists a positive tolerance $\delta_k < (u_i - \ell_i)/2$, $i = 1, \dots, n$, such that

$$\delta_k \leq \min \left\{ \frac{\varepsilon_k}{\rho_k}, \varepsilon_k \right\}$$

and, for all $i = 1, \dots, n$,

$$x^k + \delta_k e^i \in \Omega \Rightarrow x^k + \delta_k e^i \in N(x^k)$$

and

$$x^k - \delta_k e^i \in \Omega \Rightarrow x^k - \delta_k e^i \in N(x^k).$$

Theorem 2.4 Assume that $\nabla f, \nabla h, \nabla g$ satisfy Lipschitz conditions on the box Ω , defined by (19), and Assumption 2.1 holds. Then,

- The sequence $\{x^k\}$ is well defined for all k and admits limit points.
- If the sequence of penalty parameters $\{\rho_k\}$ is bounded, x^* is feasible. Otherwise, every limit point x^* is a first-order stationary point of the box constrained problem

$$\text{Minimize } \|h(x)\|_2^2 + \|g(x)_+\|_2^2 \text{ subject to } x \in \Omega.$$

- If x^* is a feasible limit point, then at least one of the following possibilities hold:

1. The point x^* fulfills the KKT conditions of

$$\text{Minimize } f(x), \text{ subject to } h(x) = 0, g(x) \leq 0, x \in \Omega.$$

2. The Cone-Continuity Constraint Qualification [2] is not satisfied at x^* for the constraints $h(x) = 0, g(x) \leq 0, x \in \Omega$.

Proof This theorem follows the proof of Theorem 2 in [24], with the exception that the Cone-Continuity Constraint Qualification (CCCQ) replaces the CPLD condition. Although CCCQ is weaker than CPLD and postdates the publication of [24], the proof remains identical as it relies solely on the fundamental properties of constraint qualifications. $\square$

3 Application to the estimation of Manning coefficients

The Saint-Venant equations [15, 37, 52, 60] describe the flow of fluids in natural channels. The state variables that describe a natural channel vary with respect to $x \in [x_{min}, x_{max}]$ (position, usually measured in meters) and $t \in [t_{min}, t_{max}]$ (time, usually measured in seconds). We define:

- $z(x, t)$: surface elevation
- $h(x, t) = z(x, t) - z_b(x)$: river depth at (x, t)
- $A(x, t)$: cross-sectional wetted area
- $Q(x, t)$: flow rate
- $P(x, t)$: wetted perimeter
- $R(x, t) = A(x, t)/P(x, t)$: hydraulic radius
- g : acceleration of gravity (approximately $9.81m/s^2$)

The Saint-Venant equations consist of two main components: Equation (20) describes mass conservation and equation (21) represents conservation of the linear momentum.

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0 \quad (20)$$

and

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x} \left(\frac{Q^2}{A} \right) + gA \frac{\partial z}{\partial x} + \frac{n_g^2 Q |Q|}{AR^{4/3}} = 0. \quad (21)$$

The Manning (also called Yen-Manning [10]) coefficient n_g represents roughness [10]. Typically, given the inlet discharge $Q(x_{min}, t)$ for $t \in [t_{min}, t_{max}]$, initial conditions $A(x, t_{min})$ and $Q(x, t_{min})$, and the coefficient n_g , numerical methods compute the values of $Q(x, t)$ and $A(x, t)$ at a set of discrete “grid points” (x_ν, t_ν) . Unfortunately, the Yen-Manning coefficient n_g is often unknown for natural channels and can vary with position x and even time t . For more details, see [10].

This research assumes that the Manning coefficient n_g varies with position x , and that a set of elevation observations $z(x_\nu, t_\nu)$ are available at n_{obs} distinct locations and times, indexed by $\nu = 1, \dots, n_{obs}$. Consequently, the spatial distribution of $n_g(x)$, discretized over a suitable grid, becomes an unknown component of our problem. Furthermore, to ensure physically plausible results, minimal variation will be required for a function $n_g(x)$ that minimizes spatial oscillations while fitting the solution of the Saint-Venant equations to the observations.

Considering a grid with n_{grid} points, distributed as $x_{\min} \leq x_1 \leq \dots \leq x_{n_{\text{grid}}} \leq x_{\max}$, the fitting problem can be formulated in several ways. For instance, in (22) the sum of squared variations is minimized, while in (23) the minimization involves the sum of absolute deviations. In (23) the total number of differing values is considered, and, in (24), the objective function is the maximum deviation between the Manning value and the auxiliary value. Although all these formulations were implemented, the analysis will focus on the results obtained with (22). Nevertheless, for completeness, some results from (22), (23), and (24) are still presented in Tables 1 and 2. The solutions produced by (22) are qualitatively similar, whereas those from (23) and (24) were noticeably inferior.

$$\text{Minimize } \sum_{i=1}^{n_{\text{grid}}} (n_g(x_i) - \bar{n})^2, \quad (22)$$

$$\text{Minimize } \sum_{i=1}^{n_{\text{grid}}} |n_g(x_i) - \bar{n}|, \quad (23)$$

$$\text{Minimize } \sum_{i=1}^{n_{\text{grid}}} H(n_g(x_i) - \bar{n}) \quad (24)$$

or

$$\text{Minimize } \max_{i=1, \dots, n_{\text{grid}}} \{|n_g(x_i) - \bar{n}|\} \quad (25)$$

where $H(t) = 1$ if $t \neq 0$ and $H(t) = 0$ otherwise. In all cases the constraint that the simulated elevations should match the observed values is required:

$$z(x_\nu, t_\nu, n_g) = z_{\text{obs}}(x_\nu, t_\nu), \quad \nu = 1, \dots, n_{\text{obs}}. \quad (26)$$

The variables of this problem are the grid-dependent Manning coefficients $n_g(x_i)$ for $i = 1, \dots, n_{\text{grid}}$, together with the scalar variable $\bar{n}$. Additionally, the model is subject to n_{obs} equality constraints (26). Moreover, additional constraints must be introduced:

$$n_g^{\text{max}} \geq n_g(x_i) \geq 0 \text{ for all } i = 1, n_{\text{grid}}. \quad (27)$$

So far, (22–27) define constrained optimization problems that have the form of (1). The variables are $n_g(x_i), i = 1, \dots, n_{\text{grid}}$ and $\bar{n}$. The constraints are given by (26), and the set Ω is defined by (27). The number of variables is $n_{\text{grid}} + 1$ and the number of constraints (excluding Ω) is n_{obs} .

It is sensible to tackle this problem using Algorithm 2.1, where the iterates x^k can be computed via different box-constrained derivative-free minimization techniques. Specifically, it is interesting to compare the traditional BOBYQA algorithm [54], introduced and implemented by M. J. D. Powell, with its modified version proposed by Ragonneau and Zhang [55]. For simplicity, the BOBYQA algorithm coded by Powell will be called BOBYQA here, whereas the BOBYQA modification [55] will be called PRIMA, which is the denomination adopted in [55]. BOBYQA has demonstrated excellent performance in numerical studies [56] even before its updating. The modifications of [55] address minor flaws and enhance BOBYQA's performance, particularly when handling cases where the objective function is ill-defined. This situation frequently arises when the function involves the numerical solution of a Partial Differential Equation (PDE), as in the case of the present research.

4 Numerical results

The numerical experiments utilize boundary data and initial conditions for a 3.3-kilometer reach of the East Fork River in Wyoming, USA, as reported in [45]. Elevation data ($z(x, t)$) from two stations ($x = 751$ m and $x = 3256$ m), recorded every 4 hours starting from May 20 (three days after the initial conditions) to June 17, form part of constraint (26). This setup results in 168 constraints per observation station, bringing the total for (26) to $n_{\text{obs}} = 336$ constraints.

To limit the Manning coefficient in constraint (27), the condition $0 \leq n_g(x_i) \leq 1$ was adopted. The differential equations were solved using the explicit finite difference method with artificial diffusion of 0.99 [27, 37, 52]. This method was chosen for its computational simplicity, considering that the equations needed to be solved many times. The spatial discretization was performed with $n_x = 101$ equally spaced points, using a time step of $\Delta t = 1s$. In this type of problem, depending on the considered coefficients, it is common to obtain undefined results (NaN). To circumvent this problem, a penalty of 1×10^{27} was applied to the objective function at these points.

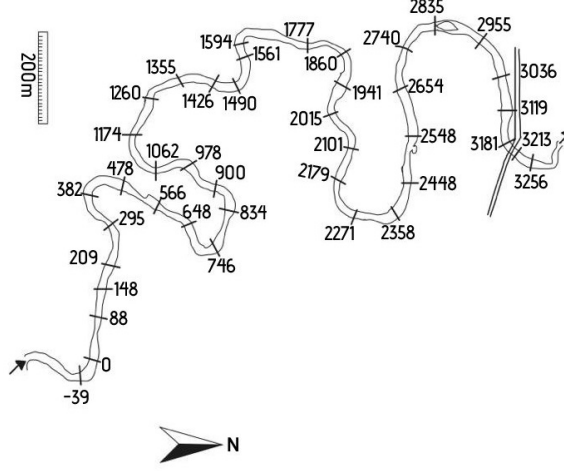


Figure 1: Illustration of the 3.3 km stretch of the East Fork River considered in the model, with markings of the data collection stations used for boundary and initial conditions, recorded between May 17 and June 17, 1977, as available in [45]. Image modified from [7].

4.1 Preliminar Comparisons

Step 2 of Algorithm 2.1 requires the minimization of an objective function subject to box constraints, thereby necessitating the selection of a specific box-constraint minimization algorithm. Since the function evaluations required during the inner iterations are computationally expensive—due to their reliance on the approximate solution of a Partial Differential Equation—it is essential to identify the most efficient solver. Therefore, various box-constraint minimization algorithms will be compared by applying them to isolated subproblems that emulate the ones found in the Augmented Lagrangian Method’s outer iterations. The algorithms demonstrating the best performance on this issue will be reserved for implementation in the full Augmented Lagrangian optimization framework, as discussed in Section 4.2.

4.1.1 The Problems

This section considers four box-constrained problems:

Problem 1

$$\text{Minimize } \sum_{i=1}^{n_{grid}} [n_g(x_i) - \bar{n}]^2 + \frac{\rho}{2} \sum_{\nu=1}^{n_{obs}} [z(x_\nu, t_\nu, n_g) - z_{obs}(x_\nu, t_\nu)]^2, \text{ subject to (27).} \quad (28)$$

Problem 2

$$\text{Minimize } \sum_{i=1}^{n_{grid}} |n_g(x_i) - \bar{n}| + \frac{\rho}{2} \sum_{\nu=1}^{n_{obs}} [z(x_\nu, t_\nu, n_g) - z_{obs}(x_\nu, t_\nu)]^2, \text{ subject to (27).} \quad (29)$$

Problem 3

$$\text{Minimize } \sum_{i=1}^{n_{grid}} H(n_g(x_i) - \bar{n}) + \frac{\rho}{2} \sum_{\nu=1}^{n_{obs}} [z(x_\nu, t_\nu, n_g) - z_{obs}(x_\nu, t_\nu)]^2, \text{ subject to (27),} \quad (30)$$

where $H(t) = 1$ if $t \neq 0$ and $H(t) = 0$ if $t = 0$.

Problem 4

$$\text{Minimize } \max_{i=1, \dots, n} \{|n_g(x_i) - \bar{n}|\} + \frac{\rho}{2} \sum_{\nu=1}^{n_{obs}} [z(x_\nu, t_\nu, n_g) - z_{obs}(x_\nu, t_\nu)]^2, \text{ subject to (27).} \quad (31)$$

For all test cases, the initial penalty parameter was set to $\rho = 5$, ensuring consistency with the first iteration of Algorithm 2.1. The spatial discretization was evaluated using four different grid resolutions, with $n_{grid} \in \{3, 15, 60, 101\}$. In cases where $n_{grid} < n_x$, the Manning coefficients are assumed to be equidistantly distributed in space, with intermediate values determined via linear interpolation. The analysis for these configurations was performed considering the observed data interval between day 3 and day 5.

4.1.2 The Algorithms

The following well established algorithms for Derivative-Free Box-Constrained Optimization were employed for solving the Problems (1–4).

1. BOBYQA [54] : This is Powell’s method for Derivative-Free Box-Constrained Optimization. The method relies on quadratic models of the objective function, which are updated at each iteration. Descent of the objective function is guaranteed through the use of a trust-region strategy.
2. PRIMA [55]: This is the implementation of BOBYQA described in [55]. The authors enhanced the original method in several respects, including addressing situations where the objective function is undefined.
3. Nelder-Mead [48]: This is likely the most traditional method used in Derivative-Free Optimization. The core mechanism involves modifying a simplex (an affinely independent set of n points) during each iteration. This is done by eliminating the worst point and generating a new trial point through the application of reflection, expansion, and contraction strategies. In this case, since the Nelder–Mead method does not handle box-constrained problems, the objective function is penalized whenever the iterate leaves the feasible region.
4. GLODS [21]: Designed specifically for Global Optimization, this algorithm uses directional searches and poll steps to explore the entire feasible region and locate promising subdomains. These promising areas are subsequently explored in detail using direct searches. By implementing multistart strategies, the algorithm collects various local minimizers, ensuring the global minimum is likely to be encountered.
5. ABC [33] : The Artificial Bee Colony (ABC) algorithm is a swarm intelligence-based metaheuristic designed for Global Optimization. It simulates the intelligent foraging behavior of honey bee swarms.
6. PSO [34]: The Particle Swarm Optimization (PSO) algorithm is a metaheuristic technique for global optimization inspired by the social behavior of bird flocking or fish schooling.
7. ECA [46]: The Evolutionary Centers Algorithm (ECA) is a physics-inspired metaheuristic designed for optimization. It draws inspiration from concepts in mechanics and physics, specifically the idea of the center of mass. ECA operates as a population-based algorithm, similar to other evolutionary techniques. The idea is that the population of candidate solutions (“particles”) is divided into groups. The center of mass is calculated for each group, and these centers act as attractors that determine the movement of the particles.

4.1.3 Algorithmic Settings

Due to the high cost of function evaluations, all algorithms were run imposing a maximum limit of 500 function evaluations. When an algorithm terminates because of its intrinsic stopping criterion, “Convergence” is reported in Table 1 and 2; otherwise, the tables reports “Exceeded.”

It should be noted that the term “convergence” can occasionally be misleading. In the context of derivative-free optimization, this diagnosis typically signifies that further improvement is unattainable from the current state. This lack of progress, often referred to as stagnation, is demonstrated differently by various methods, but it invariably relates to a predefined algorithmic small tolerance, $\varepsilon > 0$.

- In the case of BOBYQA and PRIMA, convergence is declared when the trust region radius is sufficiently small. For Nelder-Mead, convergence occurs when the size of the working simplex is small.
- The main convergence criterion in GLODS is based on the step size parameter (α) of its local directional searches. The algorithm converges when the step size (or the trust region radius, r) associated with all active points (the points representing regions of interest still under exploration) becomes sufficiently small.
- PSO, ABC, and ECA stop when the global best position has not improved by a significant amount for a specified number of consecutive iterations. This suggests the swarm has found a solution and is now circling around it.
- In addition, ECA may stop if the evolutionary centers or the entire population of particles become extremely tightly clustered. This suggests that all particles are exploring the same very small region, and further exploration is unlikely to yield better results.

The PRIMA code is written in Fortran 90, while the BOBYQA code is in Fortran 77. M. J. D. Powell originally provided the BOBYQA implementation, and Professor Zaikun Zhang provided the PRIMA code used here.

The experiments presented in this section were performed using a computer with a 5.2 GHz 12th Gen Intel(R) Core(TM) i9-12900K processor and 16 GB 3200 MHz DDR4 RAM memory, running Linux Mint 21.1. The PDE solver was implemented in Fortran 77 for use with the BOBYQA optimizer, and in Fortran 90 for experiments involving both BOBYQA and GLODS. For the metaheuristic methods, the PDE solver was reimplemented in Julia (version 1.11), using the *Metaheuristics.jl* package. In addition, we employed a MATLAB *mex* interface to call the PDE solver written in Fortran 90 within MATLAB (version R2021b) and to apply the GLODS algorithm. The codes were compiled by the GFortran compiler of GCC (version 11.3.0) with the -O3 optimization directive enabled.

4.1.4 Specific Parameter Settings

Lower and upper bounds were provided for all algorithms, allowing the variation of each $n_g(x_i)$ and $\bar{n}$ to range from 0.06 to 0.1 for all $i = 1, \dots, n_{grid}$. Other algorithmic parameters are necessary for the different algorithms:

1. For BOBYQA and PRIMA, the initial trust-region radius was set to $\rho_{beg} = 0.05$, and the final trust-region radius to $\rho_{end} = 0.001$. The number of initial interpolation points was chosen as $2n_{grid} + 4$, and the maximum number of function evaluations was set to 500.
2. For Nelder-Mead, the initial simplex size was set to 0.05 in each positive direction of the grid, and the final simplex size was defined as the largest edge length of the simplex, set to 0.001. The maximum number of function evaluations was fixed at 500.
3. For GLODS, the initial points were $n_{grid} + 1$ equidistant points in the interval $[0.06, 0.1]$. We used `search_option = 5`, which indicates that the generation of search directions is based on Sobol

sequences. The initial step size was set to `alfa_ini` = 0.05, and the initial search radius was defined as `radius_ini` = `alfa_ini`. The contraction parameter was set to `beta_par` = 0.5, and the expansion parameter to `gamma_par` = 2. The minimum tolerance was set to `tol_stop` = 0.001, and the maximum number of function evaluations was defined as `max_fevals` = 500.

4. ABC was implemented with a total of 10 generations, which, together with the population size of $N = 50$, results in a maximum of 500 function evaluations. The number of employed bees was defined as $N_e = \lfloor (N + 1)/2 \rfloor$, and the number of onlooker bees was set to $N_o = N_e$, following the standard structure of the Artificial Bee Colony method and ensuring a balanced interaction between exploration and exploitation. The abandonment limit was set to `limit` = 10, meaning that a food source is discarded if it is not improved after 10 consecutive attempts, allowing the colony to reallocate its search effort to more promising regions of the domain.
5. PSO was implemented using 10 generations with a population size of $N = 50$, resulting in a maximum of 500 function evaluations. The cognitive coefficient was set to `C1` = 2.0, which controls the influence of each particle's individual best position, while the social coefficient was defined as `C2` = 2.0, determining the influence of the global best position identified by the swarm. The inertia weight was set to $\omega = 0.8$, regulating the balance between exploration and exploitation during the velocity update.
6. In ECA, as in the other methods, the maximum number of function evaluations was 500. The parameter `eta_max` was set to 2.0, which defines the upper bound for the dynamic scaling factor controlling the exploration strength. The parameter `K` was set to 7, specifying the number of centers used to guide the search process. The exploitation probability was defined as `p_exploit` = 0.95, indicating that, at each iteration, there is a 95% chance of generating new solutions based on exploitation around the current centers. The parameter `p_bin` = 0.02 controls the probability of applying binary crossover-like perturbations to enhance diversity.

4.1.5 Preliminar Numerical Results

The numerical results corresponding to Section 4.1 are given in Table 1.

The principal findings derived from this table are presented below.

1. The number of problems in which a method achieved the lowest objective function was as follows: BOBYQA (5), PRIMA (3), Nelder-Mead (4), GLODS (3), and PSO (3).
2. Considering only PRIMA and Nelder-Mead, the count of problems where each method attained the lowest function value was PRIMA (11) and Nelder-Mead (6).
3. Considering only BOBYQA and Nelder-Mead, the count of problems where each method attained the lowest function value was BOBYQA (11) and Nelder-Mead (6).
4. Considering only BOBYQA and PRIMA, the count of problems where each method attained the lowest function value was BOBYQA (8) and PRIMA (9).
5. Considering only BOBYQA and GLODS, the count of problems where each method attained the lowest function value was BOBYQA (11) and GLODS (5).
6. Considering only PRIMA and GLODS, the count of problems where each method attained the lowest function value was PRIMA (11) and GLODS (5).

Despite being obvious, it must be warned that these numerical results are very specifically linked to the problem considered and should in no way be generalized to other problems. In particular, the PRIMA algorithm has been shown to be an advancement over BOBYQA in broader families of related problems reported in [55]

Moreover, it must be considered that algorithms like GLODS have been especially designed to discover essentially hidden global minimizers, which motivates the use of a considerable amount of computational time. Nevertheless, GLODS's performance appears to be quite reasonable in the problems considered here.

Finally, population-based algorithms like ABC, PSO, and ECA are recommended for problems with enormous and capricious oscillations, which is not the case for those studied in this paper. For these reasons, it seems that the best methods are of the BOBYQA type, in addition to Nelder-Mead, all of which are aimed at finding probable local minimizers.

4.2 Augmented Lagrangian using BOBYQA and PRIMA

In this section, the applications of BOBYQA and PRIMA to the subproblems generated in Step 2 of Algorithm 2.1 are hereafter referred to as AL-BOBYQA and AL-PRIMA, respectively. For these approaches, the following algorithmic parameters were defined: $\tau = 0.1$, $\gamma = 5$, $\lambda_{min} = -10^3$, $\lambda_{max} = \mu_{max} = 10^3$ and $\rho_1 = 5$. Each subproblem (4) associated with the Augmented Lagrangian algorithm was solved using BOBYQA or PRIMA. We set the initial and final trust region radio for BOBYQA and PRIMA to $\rho_{beg} = 0.05$ and $\rho_{end} = 0.001$, respectively. The maximum number of evaluations was set to 1000. The initial approximation was $n_g(x_i) = \bar{n} = 0.08$ for every $i = 1, \dots, n_{grid}$ with $n_{grid} = 101$. Both BOBYQA and PRIMA used $2n_{grid} + 4$ points for their quadratic interpolations. The feasibility with respect to the constraints (26) is given by the "Root Mean-Square Deviation" RMSD defined by

$$\text{RMSD} = \sqrt{\frac{1}{n_{obs}} \sum_{\nu=1}^{n_{obs}} (z(x_{\nu}, t_{\nu}, n_g) - z_{obs}(x_{\nu}, t_{\nu}))^2}.$$

For this type of problems it is not expected, in general, to obtain feasible points, which may not exist at all. Therefore, the execution of Algorithm 2.1 terminated once it completed 5 outer iterations. The results obtained using the AL-BOBYQA and AL-PRIMA solvers are reported in Table 3, while Figure 2 presents the corresponding plots. The figures display the estimated water surface elevation profiles $z(t, 751)$ and $z(t, 3256)$, together with the reference data for the period from day 3 to day 31.

Observe that the solutions obtained were satisfactory in most of the test cases. The only clearly unsatisfactory result occurs for the problem (24) subject to (26) and (27), where AL-BOBYQA fails to produce a good solution. Although the corresponding solution obtained with AL-PRIMA is not the best among all problems, AL-PRIMA is still able to provide a reasonable fit of the data without significant error, as shown in Table 3.

Figure 3 exhibits the evolution of AL-BOBYQA and AL-PRIMA showing the actual RMSD as a function of the number of function evaluations. Note that, in the initial stages, AL-BOBYQA attains smaller function values than AL-PRIMA in three of the four problems. However, for the objective function (24), AL-BOBYQA is unable to produce a satisfactory solution. Although both methods can be applied, AL-PRIMA is the only one that successfully solves all problems. This may occur because the AL-PRIMA solver corrects some of the issues reported in [55].

Although Tables 1 and 2 show that, even for n_{grid} small, reasonable results can be obtained, this is not the case when considering a longer time interval, in this case 31 days. To observe this behavior, we considered discretizations ranging from 2 to n_{grid} points, in increasing order. The results showed that the RMSD error for discretizations with fewer than 10 points was larger than 0.16 (16 cm), which is substantially higher than the errors obtained with the full discretization ($n_{grid} = 101$), approximately 6 cm. The results for $n_{grid} = 2$ are shown in Figure 4. It is also possible to observe that when n_{grid} is small, the model is unable to accurately reproduce elevations at two distinct points. In such cases, attempts to fit both points simultaneously cause the solution to become unstable and tend toward a NaN value.

Table 1: Results of the optimization methods applied to Problem 1 (sum of squared differences between each Manning coefficient and the auxiliary variable), Problem 2 (sum of the absolute differences between each Manning coefficient and the auxiliary variable), and Problem 3 (total number of Manning coefficients that differ from the auxiliary variable).

Prob.	n_{grid}	Metric	Optimization Methods						
			BOBYQA	PRIMA	N-M	GLODS	ABC	PSO	ECA
1	3	RMSD	0.028	0.029	0.029	0.271	0.270	0.264	0.230
		Objective Function	0.026	0.026	0.026	1.835	0.731	0.696	0.532
		CPU-Time(sec.)	88	82	196	161	834	192	1613
		Evaluations	58	48	114	131	500	60	500
		Stopping Criterion	Convergence	Convergence	Convergence	Convergence	Exceeded	Convergence	Exceeded
	15	RMSD	0.075	0.168	0.049	0.236	0.254	0.232	0.235
		Objective Function	0.157	0.742	0.293	1.444	0.647	0.541	0.554
		CPU-Time(sec.)	212	208	1055	616	1367	1350	1596
		Evaluations	113	98	500	504	500	400	500
		Stopping Criterion	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Convergence	Exceeded
	60	RMSD	0.148	0.159	0.181	0.215	0.178	0.123	0.174
		Objective Function	0.577	0.664	11.503	1.208	0.318	0.162	0.307
		CPU-Time(sec.)	1039	1074	1719	695	1385	1105	1365
		Evaluations	327	370	500	503	500	500	500
		Stopping Criterion	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded
	101	RMSD	0.037	0.033	0.255	0.147	0.143	0.059	0.080
		Objective Function	0.043	0.047	3.027	0.568	0.213	0.052	0.068
		CPU-Time(sec.)	719	702	647	681	677	751	619
		Evaluations	500	500	500	537	500	500	500
		Stopping Criterion	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded
2	3	RMSD	0.034	0.031	0.307	0.268	0.292	0.264	0.234
		Objective Function	0.118	0.104	0.100	1.872	0.854	0.696	0.705
		CPU-Time(sec.)	41	61	174	231	899	251	1603
		Evaluations	28	36	104	178	500	80	500
		Stopping Criterion	Convergence	Convergence	Convergence	Convergence	Exceeded	Convergence	Exceeded
	15	RMSD	0.105	0.231	0.119	0.242	0.275	0.234	0.239
		Objective Function	0.401	1.476	2.610	1.555	0.792	0.658	0.636
		CPU-Time(sec.)	185	178	797	572	1439	1439	1632
		Evaluations	98	90	368	501	500	500	500
		Stopping Criterion	Convergence	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Exceeded
	60	RMSD	0.183	0.148	0.220	0.122	0.219	0.152	0.165
		Objective Function	0.947	0.672	10.337	0.551	0.777	0.499	0.587
		CPU-Time(sec.)	976	956	1759	720	1359	1225	1363
		Evaluations	311	285	501	520	500	500	500
		Stopping Criterion	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded
	101	RMSD	0.056	0.249	0.255	0.182	0.265	0.135	0.107
		Objective Function	0.189	1.699	3.027	1.077	1.099	0.904	0.841
		CPU-Time(sec.)	719	480	648	604	585	579	619
		Evaluations	500	413	501	503	500	500	500
		Stopping Criterion	Exceeded	Convergence	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded
3	3	RMSD	0.148	0.142	0.151	0.265	0.268	0.264	0.230
		Objective Function	3.569	3.524	3.593	1.833	3.719	3.696	3.530
		CPU-Time(sec.)	33	36	84	163	1619	193	1604
		Evaluations	24	21	49	130	502	60	500
		Stopping Criterion	Convergence	Convergence	Convergence	Convergence	Exceeded	Convergence	Exceeded
	15	RMSD	0.304	0.162	0.302	0.237	0.276	0.232	0.233
		Objective Function	5.408	0.685	17.380	1.454	15.760	10.540	15.541
		CPU-Time(sec.)	101	192	284	621	1476	1537	1652
		Evaluations	56	96	130	509	500	500	500
		Stopping Criterion	Convergence	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Exceeded
	60	RMSD	0.305	0.212	0.303	0.202	0.231	0.240	0.181
		Objective Function	5.426	1.173	62.382	1.061	60.534	26.575	60.329
		CPU-Time(sec.)	445	918	1588	691	1370	1645	1256
		Evaluations	146	318	447	500	501	500	500
		Stopping Criterion	Convergence	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Exceeded
	101	RMSD	0.302	0.241	0.303	0.148	0.151	0.333	0.131
		Objective Function	11.378	1.512	3.382	0.572	101.227	45.107	101.171
		CPU-Time(sec.)	389	511	704	633	583	657	578
		Evaluations	284	457	501	528	500	500	500
		Stopping Criterion	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded

Table 2: Results of the optimization methods applied to Problem 4 (maximum difference between the Manning coefficients and the auxiliary variable).

Prob.	n_{grid}	Metric	Optimization Methods						
			BOBYQA	PRIMA	N-M	GLODS	ABC	PSO	ECA
4	2	RMSD	0.140	0.131	0.062	0.278	0.269	0.264	0.230
		Objective Function	0.561	0.491	0.199	2.020	0.738	0.696	0.537
		CPU-Time(sec.)	13	26	132	229	850	150	1032
		Evaluations	18	20	99	179	500	70	500
		Stopping Criterion	Convergence	Convergence	Convergence	Convergence	Exceeded	Convergence	Exceeded
	15	RMSD	0.177	0.168	0.027	0.242	0.253	0.230	0.236
		Objective Function	0.833	0.757	0.092	1.537	0.653	0.550	0.572
		CPU-Time(sec.)	161	98	755	642	1487	1525	1610
		Evaluations	88	200	392	500	502	450	500
		Stopping Criterion	Convergence	Convergence	Algorithm	Exceeded	Exceeded	Convergence	Exceeded
	60	RMSD	0.132	0.154	0.098	0.232	0.181	0.138	0.165
		Objective Function	0.469	0.669	0.326	1.412	0.337	0.210	0.281
		CPU-Time(sec.)	1510	1168	1682	702	1329	968	1337
		Evaluations	500	358	501	505	502	500	500
		Stopping Criterion	Convergence	Convergence	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded
	101	RMSD	0.019	0.037	0.279	0.108	0.185	0.067	0.075
		Objective Function	0.019	0.046	2.095	0.317	0.356	0.065	0.066
		CPU-Time(sec.)	670	696	558	628	581	690	571
		Evaluations	500	500	500	519	504	500	500
		Stopping Criterion	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded	Exceeded

Table 3: RMSD values obtained by AL-BOBYQA and AL-PRIMA for the general problems defined by the objective function (22) (sum of squared errors), (23) (sum of absolute deviations), (24) (total number of differing values), and (25) (maximum deviation between the Manning value and the auxiliary value), subject to (26) and (27) with $n_{grid} = 101$.

Method	(22) subject to (26)-(27)		(23) subject to (26)-(27)		(24) subject to (26)-(27)		(25) subject to (26)-(27)	
	RMSD	Eval.	RMSD	Eval.	RMSD	Eval.	RMSD	Eval.
AL-BOBYQA	0.054	3218	0.055	4129	0.244	2134	0.051	4164
AL-PRIMA	0.052	3246	0.053	3304	0.076	2333	0.052	3177

4.3 Comparison with alternative approaches

Well-established methods for estimating Manning coefficients from empirical elevation observations are based on minimizing the discrepancies between observed elevations and those estimated by solving the Saint-Venant equations. Consequently, the objective function to be minimized depends solely on the Manning coefficients. Generally, it is assumed that the Manning coefficient remains constant along the river course and over time; alternatively, it has been assumed that well-defined spatio-temporal regions where the coefficient is constant can be determined based on physical considerations [7, 10]. In this way, the error to be minimized depends on only a few variables. To obtain a plausible approximation of the Saint-Venant model to the observations, Ayvaz [7] proposes spatial jumps in the Manning’s coefficient at predetermined positions, while Birgin et al [10] establish temporal jumps in this coefficient at specific times, based on meteorological conditions. The novelty of our approach lies in the fact that we account for the variability of the Manning coefficient without requiring additional physical hypotheses to justify its spatial changes. To demonstrate that this approach outperforms well-established ones, we will present a comparison between our results and those obtained by assuming a constant Manning coefficient.

4.3.1 Constant Manning approach

In Table 4, we present the results of estimating the Manning coefficient using elevation data from two different stations on the Fork River, following the algorithm introduced in [10]. We report the number of calls to the subroutine that solves the Saint-Venant equations. Note that, within the context of our Augmented Lagrangian approach, the model based on the assumption of zero Manning variation is essentially equivalent to the penalized formulation for ρ close to zero. Given that this is a restrictive assumption, high accuracy in the reproduction of observations is not expected.

Initial and final day	Manning Coefficient	RMSD	Calls to SV	CPU-time (seconds)
0 – 30	9.099 E-2	0.2012	72	377.0
3 – 30	9.099 E-2	0.2019	72	336.8
0 – 10	0.1122	0.1982	66	121.2
3 – 10	0.1157	0.1867	78	99.51
10 – 20	9.099 E-2	0.1953	72	120.6
20 – 30	0.1141	0.1395	78	145.6

Table 4: Estimation of only one Manning Coefficients using the approach of [10] with precision 10^{-4} . Fork River, $n_{grid} = 101$. Initial estimation: 0.08. Observations at $x = 751$ and $x = 3256$.

4.3.2 Detailed Analysis of the Augmented Lagrangian approach

The analysis of the application of Algorithm 2.1 to solve problem (22) subject to (26) and (27), using the parameters defined at the beginning of this section, follows. The results are given in Table 5.

It can be observed that the first two iterations of the algorithm play a key role in driving the solution toward an approximately feasible point, promoting a significant reduction in the constraint violation. From the third iteration onward, the method mainly focuses on refining the previously obtained solution, leading to more subtle adjustments of the decision variables. Moreover, the value of the variational objective function (22) varies across the iterations, reflecting the inherent trade-off of the Augmented Lagrangian method between data fitting and constraint enforcement. In particular, the increase observed in the objective function at later iterations indicates that the solution converges to a non-uniform spatial profile of the Manning coefficient. The progressive update of the penalty parameter throughout the iterations strengthens the enforcement of the constraints, contributing to the computation of increasingly feasible solutions for the problem under consideration.

Table 5: Augmented Lagrangian method for solving (22) subject to (26) and (27). MV = Manning Variation, QE = Quadratic Error. Total number of evaluations: 2686.

Outer Iteration	Evaluations	Time (s)	MV	QE	L_{ρ_k}	RMSD
0	0	0	0.0000	25.0414	25.0414	0.2722
1	613	4508	0.0078	12.0092	30.0309	0.1885
2	739	6114	0.0262	1.3110	17.2486	0.0623
3	681	9278	0.0347	0.9708	73.1176	0.0536
4	653	5647	0.0418	0.9371	346.3779	0.0527

4.3.3 Comparison with LOG-DS

The LOG-DS method for derivative-free optimization with inequality constraints [13] is a modification of the approach introduced in [14]. This version employs a direct search approach to solve the subproblems. Reference [13] describes both the method and its corresponding software, providing numerical evidence that it performs at least as well as more widely known alternatives. Furthermore, the software is easily accessible, and the authors provide prompt and helpful assistance with implementation issues. For these reasons, LOG-DS was selected for comparison with the new version of the Augmented Lagrangian method using BOBYQA or PRIMA. This comparison is restricted to the specific problem of interest and does not extend to other constrained optimization problems.

In the problem considered in this paper, the number of constraints exceeds the number of variables, a configuration that typically poses numerical challenges for LOG-DS [13]. In order to improve the behavior of LOG-DS, we reformulated the constraints (26) to an inequality-based structure, as follows:

$$|z(x_\nu, t_\nu, n_g) - z_{obs}(x_\nu, t_\nu)| - 0.01 \leq 0, \quad \nu = 1, \dots, n_{obs}, \quad (32)$$

The algorithmic parameters employed by LOG-DS are summarized in Table 6.

Table 6: Algorithmic parameters for LOG-DS

Parameter	Value	Parameter	Value
obj.tol_alfa	10^{-8}	obj.epsilon_ini	10^{-2}
obj.fevals_max	4000	obj.tol_feasible	10^{-5}
obj.tol_grad	10^{-5}	obj.eta	0.35
obj.iter_max	4000	obj.v	1.1
obj.theta	0.5	obj.beta_par	1.0
obj.phi	1.0	obj.zeta_par	10^2
obj.rho_in	10^{-3}	obj.gamma_par	10^{-5}

The proposed methodology was applied to the full-scale problem, considering a spatially distributed Manning coefficient across the computational grid. With a discretization of $n_{grid} = 101$ points and a dataset spanning 28 days (from day 3 to day 31) with observations recorded every 4 hours, the formulation resulted in $n_{obs} = 336$ constraints. The Manning coefficient was bounded within the physical range $0 \leq n_g(x_i) \leq 1$ for all $i = 1, \dots, n_{grid}$.

Under these conditions, the LOG-DS algorithm reached the limit of 4,000 function evaluations, yielding a Manning Variation (MV) of 0.2769, a Quadratic Error (QE) of 0.7464, and a Root Mean Square Deviation (RMSD) of 0.0470. The resulting hydrographs and the comparison with observed data are illustrated in Figure 5.

To benchmark the performance of the proposed algorithm against the methods described in Section 2, specifically AL-BOBYQA and AL-PRIMA, we analyze the final objective function (Manning Variation MV) values, feasibility, and the total number of function evaluations. The comparative results for each approach are summarized in Table 7.

Table 7: Comparison of optimization results across AL-BOBYQA, AL-PRIMA, and LOG-DS.

Metric	AL-BOBYQA	AL-PRIMA	LOG-DS
Objective Function (MV)	0.0981	0.0418	0.2749
QE	0.8967	0.8906	0.7464
RMSD	0.0515	0.0527	0.0470
Function Evaluations	3341	2688	4000

The convergence behavior is further illustrated in Figures 6 and 7. Figure 6 presents the evolution of the objective function MV relative to the number of evaluations, while Figure 7 displays the reduction of the RMSD (error in elevation recovery) as the optimization progresses.

It can be observed that all methods tested in this study successfully solved the problem with low error margins. However, each approach exhibits distinct performance characteristics. The LOG-DS algorithm achieved better values for the RMSD, which reflects the infeasibility of the constraints. Indeed, the infeasibility error for LOG-DS was 4.7 cm, while the RMSD for the AL-BOBYQA and AL-PRIMA methods reached 5.2 cm. From a practical standpoint, these errors are negligible. On the other hand, the Manning-Variation (MV, the objective function in (22)) was six times greater than that obtained by AL-BOBYQA and AL-PRIMA. See Figure 8.

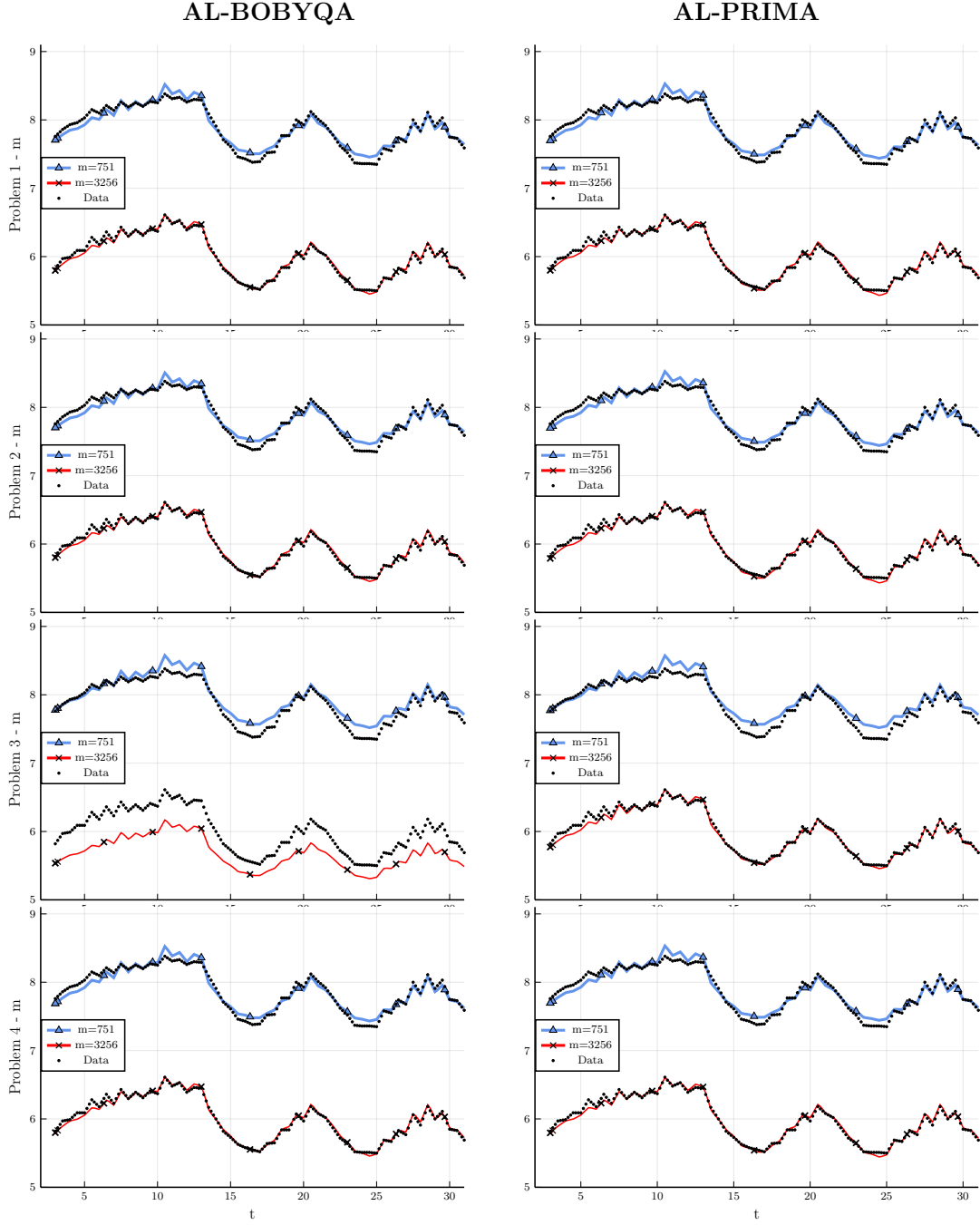


Figure 2: Hydrograms at positions $x = 751$ meters and $x = 3256$ meters comparing the results of the Problems (22–25) subject to (26) and (27) using Algorithm 2.1 with BOBYQA or PRIMA for handling subproblems with $n_{grid} = 101$. Observed data ($n_{obs} = 336$) are represented by circular markers. The vertical axis represents the elevation $z(x, t)$ in meters, while the horizontal axis corresponds to time t in days.

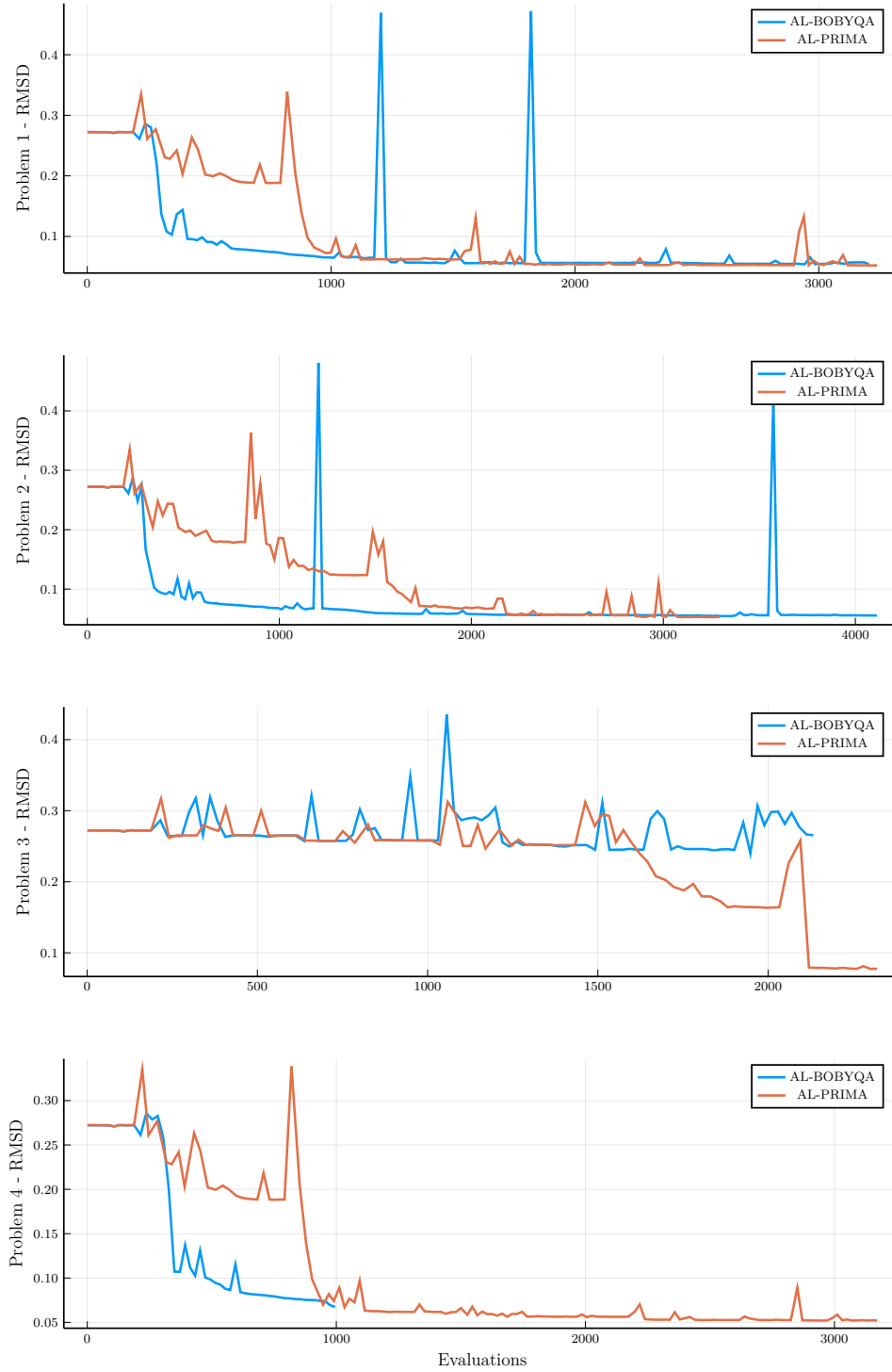


Figure 3: AL-BOBYQA versus AL-PRIMA: Evolution of the RMSD (error in recovering the elevation observations) as a function of the number of function evaluations for Problems (22–25) subject to (26) and (27) with $n_{grid} = 101$. Blue line represents the AL-BOBYQA results, while orange line represents the AL-PRIMA results.

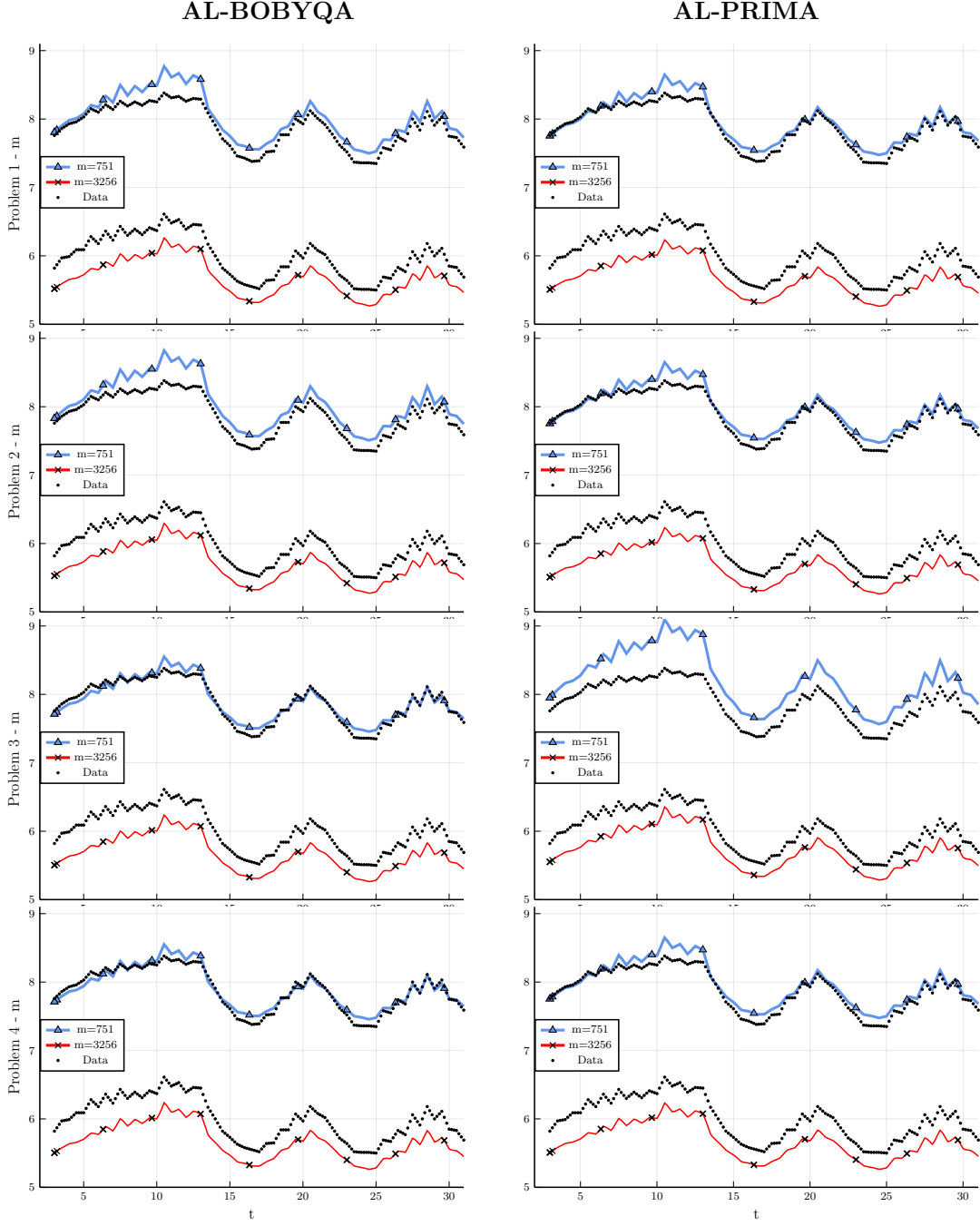


Figure 4: AL-BOBYQA and AL-PRIMA for the problem (22–25) with $n_{grid} = 2$. Observe poor recovery of hydrograms at positions $x = 751$ meters and $x = 3256$ meters. Observed data ($n_{obs} = 336$) are represented by circular markers. The vertical axis represents the elevation $z(x, t)$ in meters, while the horizontal axis corresponds to time t in days.

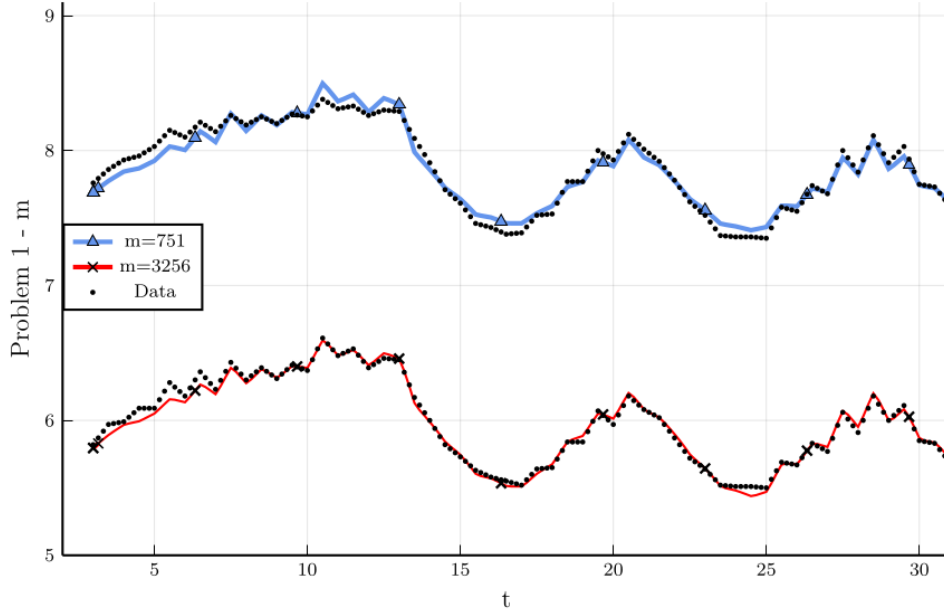


Figure 5: Hydrographs at positions $x = 751$ m and $x = 3256$ m comparing the results of the optimization problem defined by the objective function (22) with the constraints (32–27) using the LOG-DS algorithm. Observed data ($n_{obs} = 336$) are indicated by circular markers. The vertical axis represents the water surface elevation $z(x, t)$ in meters, while the horizontal axis denotes time t in days.

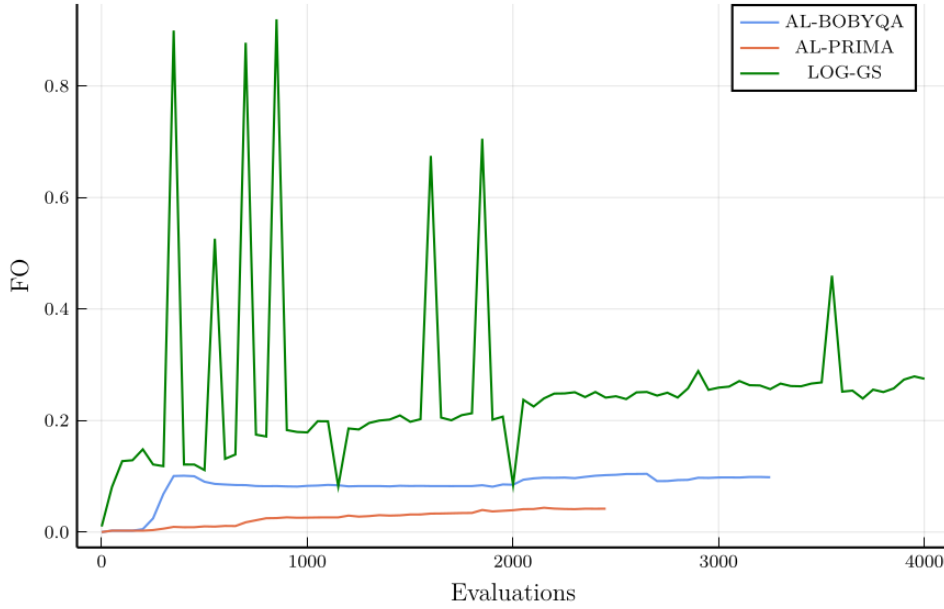


Figure 6: AL-BOBYQA vs. AL-PRIMA vs. LOG-DS: Evolution of the objective function (22) as a function of the number of evaluations. Constraints follow (26) for AL-BOBYQA/AL-PRIMA and (32) for LOG-DS. The blue line represents AL-BOBYQA, the orange line represents AL-PRIMA, and the green line represents LOG-DS.

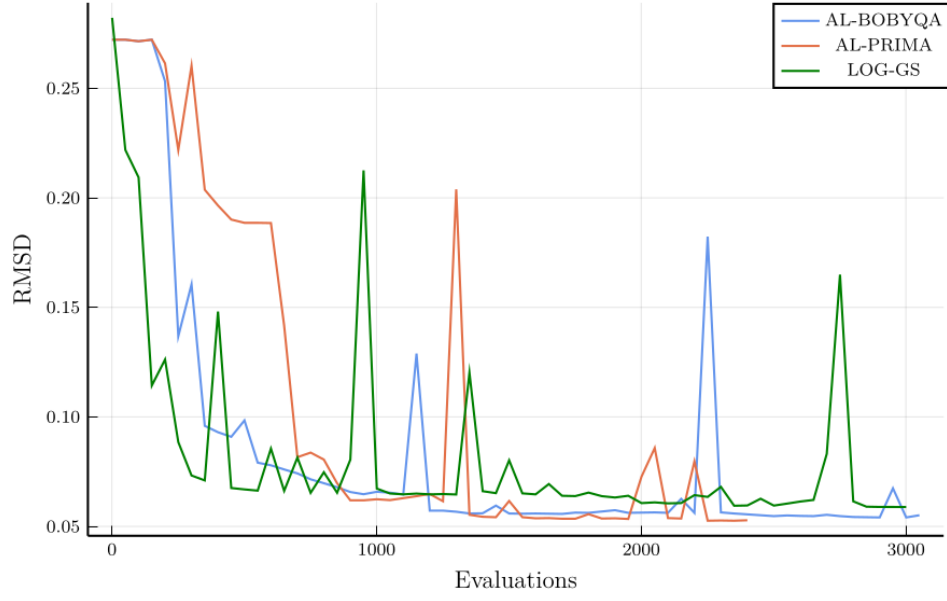


Figure 7: AL-BOBYQA vs. AL-PRIMA vs. LOG-DS: Evolution of the RMSD (error in elevation recovery) as a function of the number of evaluations. Problems (22) subject to (26–27) (AL-BOBYQA/AL-PRIMA) and (22) subject to (32–27) (LOG-DS). The blue, orange, and green lines represent AL-BOBYQA, AL-PRIMA, and LOG-DS, respectively.

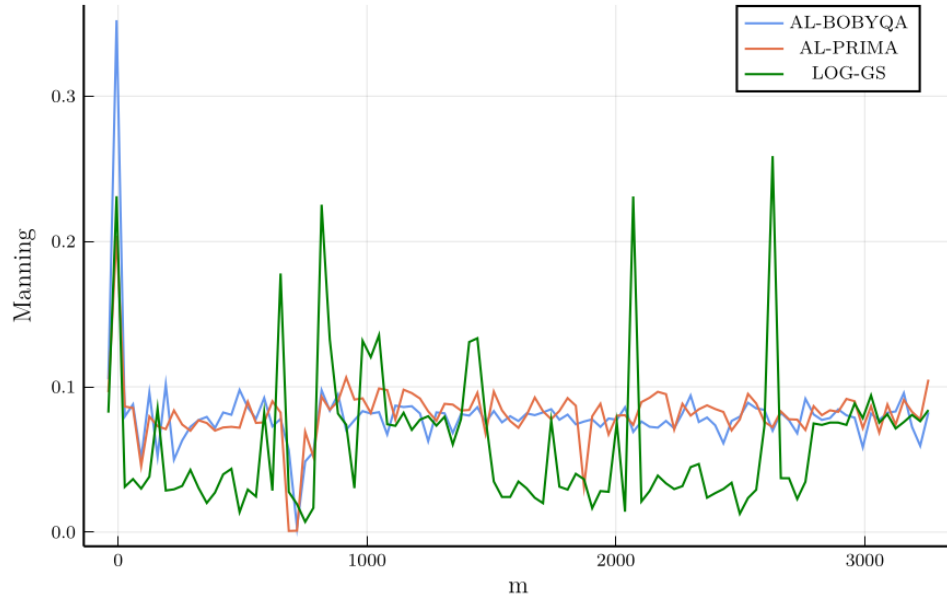


Figure 8: Manning Coefficient solution curves $n_g(x)$ computed by AL-BOBYQA, AL-PRIMA, and LOG-DS: The blue, orange, and green lines represent AL-BOBYQA, AL-PRIMA, and LOG-DS, respectively.

5 Conclusions

This paper presented a new version of the Safeguarded Augmented Lagrangian method without derivatives. This formulation appears to be well-suited for cases where Derivative-Free methods are used to solve the Augmented Lagrangian subproblems. This is because Derivative-Free methods typically declare convergence when the current iterate exhibits a smaller function value than its neighbors. To address this, set $N(x^k)$ was introduced, which comprises points with larger function values than x^k . Consequently, the neighbors used to terminate subproblem solver executions form a subset of $N(x^k)$. As a result, arguments relying on the differentiability of the objective function or constraints are not employed at all.

The new method is applied to a novel formulation of the problem of estimating Manning coefficients of natural rivers based on the Saint-Venant equations. The novelty lies in the fact that, in nature, Manning coefficients are not constant, but it is sensible to impose a condition of minimal variation. This formulation gives rise to a constrained optimization problem, whose solution is obtained using the Safeguarded Augmented Lagrangian Method together with several derivative-free solvers, including the classical BOBYQA algorithm of Powell and its updated version, PRIMA, as well as the Nelder-Mead method, GLODS, ABC, ECA, and PSO. In our problem, the differences between BOBYQA and PRIMA were not significant, and both solvers outperformed the other methods in preliminar numerical experiments concerning the solution of Augmented Lagrangian subproblems.

The LOG-DS algorithm [13] has proven to be at least as efficient as many well-established software packages for derivative-free constrained optimization. This fact, in addition to its user-friendliness regarding installation and efficient maintenance, was decisive in planning its comparison with the new algorithms for the hydraulic problems considered in this paper. The results revealed that LOG-DS and the Augmented Lagrangian methods achieve reproductions of the elevations with similar precision, with a slight advantage for LOG-DS. On the other hand, the profile for the Manning coefficients produced by LOG-DS (figure 8) falls outside the usually acceptable limits for this type of problem, whereas the variation in the Manning coefficients computed by AL-PRIMA and AL-BOBYQA is plausible and consistent with the literature.

Augmented Lagrangian methods naturally benefit from advances in box-constrained and linearly-constrained optimization. Since the publication of [18] in 2009, significant progress has been made in unconstrained, box-constrained, and linearly constrained optimization, as seen in works such as [5, 24, 49, 26, 50, 51], among others. The Augmented Lagrangian approach is particularly well-positioned to leverage these advances, providing a versatile framework for constrained optimization. The integration of recent techniques, such as full-low evaluation [8, 59] and probabilistic descent [29, 30, 57], into the Augmented Lagrangian framework warrants further exploration and testing for solving subproblems. Furthermore, the increasing tractability of complex subproblem structures, made possible by recent research [9, 19, 22], presents opportunities for developing practical and applicable software by incorporating these subproblems into Augmented Lagrangian frameworks. In particular, multiobjective and equilibrium problems [1, 20, 47] with intricate constraints represent a promising application area, deserving special attention and further investigation.

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